

Week8_group&indivisual_assignment__Sulaiman (Stan) Abu Romman

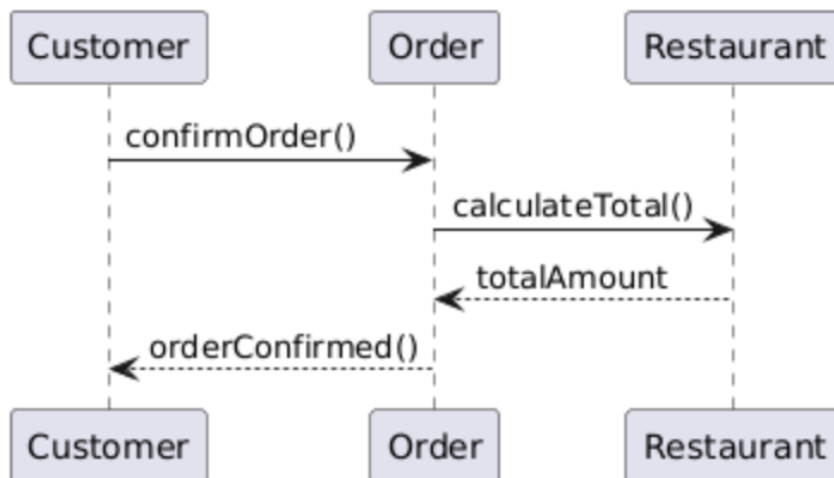
Part 1 – Sequence Diagrams (SD)

A Sequence Diagram is an interaction diagram used in the design phase of software development. It shows how objects within a system communicate with each other over time by sending messages. Unlike a System Sequence Diagram, which treats the system as a black box, a Sequence Diagram shows the internal objects and their interactions.

Sequence Diagrams are created based on Operation Contracts. Each operation defined in an Operation Contract is expanded into a Sequence Diagram that shows how the operation is executed internally.

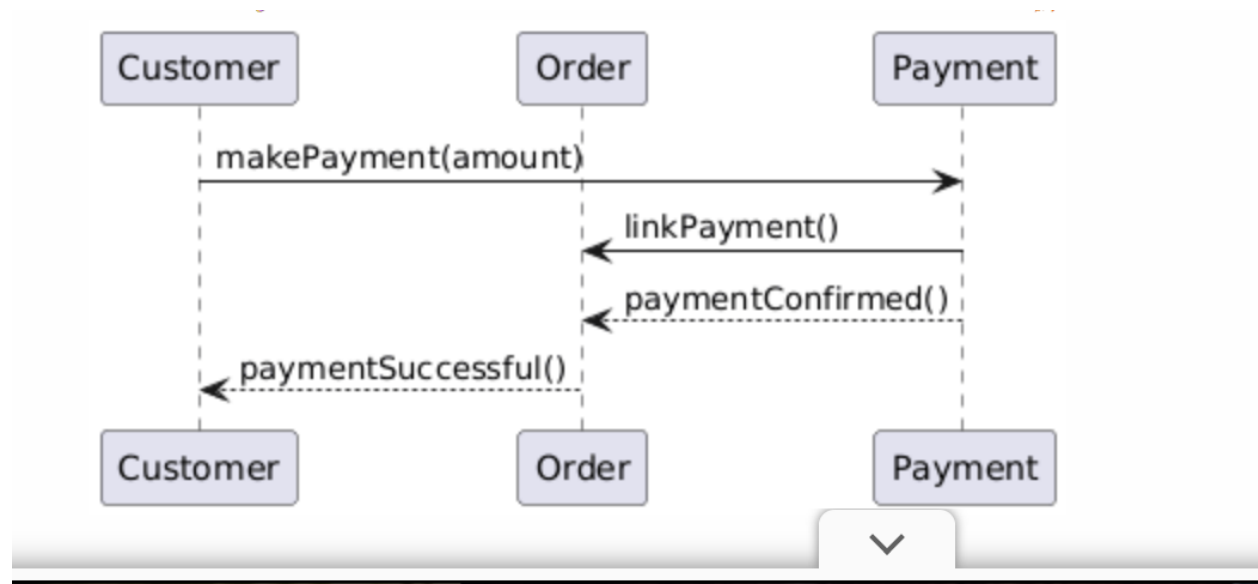
SD 1 – confirmOrder() – Stomach Saver

This sequence diagram represents how the system internally handles the confirmOrder operation in the Stomach Saver system. It shows how objects collaborate to confirm an order and calculate the total amount.



SD 2 – makePayment(amount) – Stomach Saver

This sequence diagram illustrates the internal execution of the makePayment operation. It shows how the payment is created, linked to the order, and confirmed.



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A Design Class Diagram (DCD) is a diagram that shows the classes of a system during the design phase, including their attributes, methods, and relationships. Unlike a domain model, which focuses on conceptual objects, a DCD is closer to implementation and includes method signatures derived from sequence diagrams.

A Design Class Diagram is directly linked to Sequence Diagrams. The messages shown in sequence diagrams become methods in the classes of the DCD. Each object interaction in an SD helps identify responsibilities and operations that must be implemented.

The purpose of a DCD is to translate system behavior into a structured class design that can be implemented in code. It helps developers understand class responsibilities, method definitions, and relationships before programming begins.

Draft Design Class Diagram – Stomach Saver

The following draft Design Class Diagram represents the main classes involved in the Stomach Saver system. The diagram includes classes such as Customer, Order, Payment, and Restaurant, along with their key attributes and methods derived from the sequence diagrams.

